Muhammad Naufil

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Education

Saarland University April 2024 - Present
MS Computer Science Saarbrücken, Germany
Master Thesis at **MPI-Informatics**: Group Co-Speech Gesture Synthesis

- **Built audio speech and text-conditioned 3D human gesture generation model** using latent diffusion (a Transformer DiT denoising in the latent space), conditioned on T5 text and mel-spectrogram audio features via cross-attention, generating full-body SMPL-X motion (body, fingers, locomotion) at scale on the BEAT2, embody3d and DnD Dataset
- **Addressed research questions**: How to make characters spatially and socially aware? How to prevent collisions and ensure appropriate positioning and orientation for mutual gaze? How to efficiently encode multi-participant speech and audio without excessive computational cost? How to design a scalable architecture for a variable number of participants?

NED University of Engineering and Technology Dec 2016 - Oct 2020
Bachelor of Electronic Engineering, CGPA 3.812/4 Karachi, Pakistan
Bachelor Thesis: Self-Driving Car, predicts steering angle, avoids obstacles and classifies traffic signal lights using Jetson Nano

Experience

Mercedes August 2026 - Present
Werkstudent Stuttgart, Germany

- Developing data processing pipelines to transform raw camera and LiDAR ROS bag recordings into structured, ML-ready datasets for autonomous driving model training.

University of Saarland July 2026 - Present
Research Assistant Saarbrücken, Germany

- Understanding sign language and co-speech gestures through phonemes

Detesia - CISPA Startup Oct 2024 - Dec 2025
Computer Vision Working Student - Deepfake Detection Saarbrücken, Germany

- Finetuning vision foundation models on a huge collection of deepfakes with access to clusters of 8xA100 GPUs.
- Large-scale generation of deepfakes using state-of-the-art generative open and closed source models
- Learned that features distribution for every generative model is widly different and thus requires us finetune our deepfake detection model for every new generative model that comes out to learn its unique distribution

Gordian Robotics - Retail AI April 2024 - June 2024
Founding Engineer Remote

- Developed a one-shot computer vision system for retail shelf monitoring; evaluated SuperGlue and LightGlue to recognize products from a single reference image and detect out-of-stock items without retraining for frequent packaging changes.

Retrocausal, Inc. Nov 2020 - March 2024
Computer Vision Engineer Remote

3D Human Pose Estimation:

- Created markerless MoCap product with a single collague for a $150k yearly contract with **Siemens**.
- Landed **3M** as the first customer for the company.
- Experimented with monocular and multi-view approaches to extract 3D joint locations of factory floor workers.
- Addressed monocular ambiguity issues: limb length constraints, temporal smoothing.
- Synthesised human motion in a high-quality anthropomorphic simulation to allow detailed ergonomics studies.
- Extended open-source monocular synthetic dataset of 3D motion generation pipeline to multi-view using Blender, while incorporating camera intrinsics, extrinsics, world coordinates and camera coordinates.
- Created a mini motion capture (MoCap) system using two RGB-D cameras with a confidence-based fusion approach.

Human Action Recognition / Segmentation:

- Created the human activity segmentation demo for the SBIR grant from **NASA's Human Research Program (HRP)**.
- Reduced purchasing price by 150x and deployment time by 180x for a discrete manufacturing assembly customer by replacing their sensor and visual inspection system with a human action recognition model deployed at Jetson Xavier, Jetson AGX
- Wrote an algorithm to perform time and motion studies on discrete manufacturing assembly processes to provide step-level analytics.
- Ran pilot projects with **Honda, Suzuki, Nissan**, from supervising data collection/annotation team to iteratively improve models to adapt to diverse scenarios on the factory floor
- Evaluated video understanding methods (3D CNN, I3D) against simple object detection (YOLO) and realized that simple object detection outperforms for detecting assembly procedures on the factory floor

Technical Skills

Programming Languages: Python, C++, JavaScript
Frameworks & Libraries: PyTorch, Diffusion Models, Vision Transformers, OpenCV
Tools & Cloud: Docker, Git, N8n, CrewAI, Openclaw, Blender, Linux, Pinecone
Interests: Computer Vision, Generative AI, Multimodal LLMs, Agentic AI, 3D Human Pose Estimation